

Troubleshooting

Troubleshooting & FAQ

This guide covers the most common symptoms users hit in production and the fastest fix path for each. Sections are grouped by what the user sees, not by component.

Notifications not arriving

Symptoms

- You're on site but didn't get an alert for a P3+ job in your trade
- Other technicians did get it
- Manager dashboards show the job was created and alerts were sent

Check, in order

1. **Notifications permission** — Settings → Apps → CTP Job Cards → Notifications → "Allowed".
On Android 13+, the app must hold `POST_NOTIFICATIONS`.
2. **Battery optimization** — Settings → Apps → CTP Job Cards → Battery → Unrestricted.
Aggressive battery savers (Samsung, Xiaomi, Huawei) kill background FCM listeners after a few minutes idle. This is the #1 cause.
3. **DND bypass** — for P4/P5 to break through Do Not Disturb, the app needs DND access.
Settings → Sound & Vibration → Do Not Disturb → App exceptions.
4. **FCM token registered** — open Settings inside the app and tap "Refresh FCM Token". If the token is null or stale, the Cloud Function has no way to reach you.
5. **On-site status** — geofencing only alerts on-site technicians. If you're showing as off-site when you're physically on site, see "Geofence not triggering" below.

If all of those check out, ask Admin to look at the `notifications` Firestore collection — it logs every send attempt and lists exactly which clock numbers were targeted.

Full-screen P5 alarm not firing

Symptoms

- P5 jobs arrive as a regular banner instead of a screen-takeover with loud alarm
- The phone is on silent or DND when the alert should bypass

Check, in order

1. **"Display over other apps" / System Alert Window** — Settings → Apps → CTP Job Cards → Special access → Display over other apps → Allow. Without this, the lock-screen takeover falls back to a banner.
 2. **Schedule exact alarms** — Settings → Apps → Special App Access → Alarms & Reminders → CTP Job Cards → Allow. P5 uses the exact-alarm API; without it the alarm fires late or not at all.
 3. **DND bypass** — see Notifications section.
 4. **Lock screen notifications** — Settings → Notifications → Lock screen → Show all content. Some OEMs hide content by default.
 5. **Sound profile** — even with bypass granted, some firmware mutes alarms in DND. Test by setting DND on and running the in-app "Test P5 Full Screen Alert" button on the Permissions onboarding page.
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Geofence not triggering

Symptoms

- Walking onto site doesn't flip you to on-site
- You never receive job alerts even with all other permissions granted

Check, in order

1. **Location permission level** — must be "Allow All the Time", not "Only while using the app". Background location is the *only* thing that triggers the geofence when the screen is off.
2. **Location services on** — phone-wide location switch must be enabled.
3. **Geofence radius** — the active geofence is ~800 m around the configured centre. If you're outside it, you're correctly off-site; the radius can be reconfigured by Admin via the **Geofence**

Editor screen.

4. **Re-registration after reboot** — Android removes geofences on device reboot. The app re-registers on next launch; if you rebooted and didn't open the app, do so once.
5. **Force-refresh location** — Settings → "Check Current Location" button. This calls `LocationService.checkCurrentLocation()` and updates `isOnSite` immediately.

If the geofence still doesn't fire, Admin can check the device's reported lat/lng in the `employees` document — that confirms whether the OS is even sending location updates.

Job cards not syncing

Symptoms

- You created a job but it doesn't appear in View Jobs
- Closure note didn't save after going offline

Check, in order

1. **Connectivity indicator** — top of the Home screen shows a sync state badge. Yellow/red means writes are queued.
2. **Sync queue contents** — Settings → Diagnostics → "View Sync Queue" (admin/dev only) lists pending writes in the Hive `sync_queue` box.
3. **Force-sync action** — tap the sync badge to manually flush. `SyncService` listens to `connectivity_plus` and replays on reconnect, but the manual trigger is faster than waiting.
4. **Firebase Auth session** — if your session expired, writes will fail silently and stay queued. Sign out and sign back in.

If writes stay queued indefinitely, check Firestore security rules (Admin) — a rule rejection looks like a network error from the client.

"Access denied. Manager role required"

Symptom

- Dashboard tab shows the access-denied message instead of the dashboard

Cause

Role is **not** a stored field — it's inferred from `Employee.position`. The Dashboard tab is gated by `position.toLowerCase().contains('manager')`. If your position string doesn't contain the word "manager" (case-insensitive), you're treated as a non-manager.

Fix

Ask Admin to update your `position` field in the `employees` Firestore collection. Common examples that work:

- "Production Manager" ✓
- "Workshop Manager" ✓
- "Manager - Electrical" ✓

Examples that **don't** work even though they look manager-ish:

- "Supervisor" ✗ (no "manager" substring)
- "Foreman" ✗
- "Team Lead" ✗

If your role is in this grey area, talk to Admin about whether to extend `lib/utils/role.dart` to recognise it.

Login fails / "No employee profile found"

Symptoms

- Firebase Auth accepts the password
- App immediately shows "No employee profile found. Please register first."

Cause

The `employees` collection is queried by `uid` (Firebase Auth UID). If your employee document exists but has no `uid` field, or has the wrong `uid`, the lookup fails.

Fix

Admin needs to set the `uid` field on your `employees/{clockNo}` document to match your Firebase Auth UID. The Firebase Console → Authentication tab shows the UID for each user account.

If the document doesn't exist at all, use the Registration screen to create one.

Copper tab missing

Symptom

- Some users see a "Copper" tab in the bottom nav, you don't

Cause

Copper Dashboard is whitelisted to clock numbers `22`, `5421`, and `20` only (see `lib/utils/role.dart`).

Fix

If you need access, ask Admin to either extend the whitelist (code change) or to use one of the existing whitelisted accounts.

Admin screen — "Incorrect password"

Symptom

- Tapping Admin in Settings prompts for a password and rejects what you type

Cause

The Admin password is a hardcoded check in `home_screen.dart` (search for `correctPassword`). It is not stored in Firestore.

Fix

Get the current password from another Admin user. If everyone has forgotten it, a developer needs to change the constant in the source and ship a new build.

Escalation didn't reach me even though I'm a manager

Symptoms

- A job hit Stage 2 escalation and the dashboard logged it
- You're on-site and have the right position, but you didn't get an alert

Check, in order

1. `notification_configs/global.stages.stage2.enabled === true` — Admin can check this in the Admin → Escalation Config tab.
 2. **Stage 2 recipient rules include your role** — typical Stage 2 rules: `onsite_dept_managers`, `onsite_workshop_manager`. If you're a general manager and only dept managers are listed, you won't be notified.
 3. **enabled_at is in the past** — if Admin recently changed Stage 2's recipients, the function skips jobs created before `enabled_at` so the new recipients aren't blasted with backlog.
 4. **You're recorded as on-site** — recipient rules with the `onsite_` prefix filter by `isOnSite == true`. Geofence section above.
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Operator follow-up alerts

What they look like

"No response yet — Job #123. 10 minutes passed with no assignment. We've notified 4 people. Follow up directly."

What they mean

These are sent to the operator who created the job, at each escalation stage, *regardless* of whether the operator is on-site. They are informational — the operator doesn't need to do anything in the app, but they can phone or radio someone directly if the job is critical.

Disabling for a specific job type

Add the job type to `excluded_job_types` in `notification_configs/global`. The Cloud Function skips initial notification *and* all escalation stages for excluded types. Common example:

```
"maintenance".
```

Where to look when nothing else works

- `notifications` **Firestore collection** — every notification ever sent, with `sentTo`, `level`, `priority`, and timestamp. Authoritative log.
- `job_card_audit` **Firestore collection** — append-only log of every job card mutation, with who and when.
- **Firestore Console** → **Functions** → **Logs** — `escalateNotifications` logs each stage's query results; useful for tracing why a job did or didn't escalate.
- `memory-bank/learnings.md` (if present) — running list of gotchas the team has hit before.
- **Diagnostics screen inside the app** — surfaces FCM token, on-site status, last geofence event, and sync queue size in one place.